



ABDOMEN

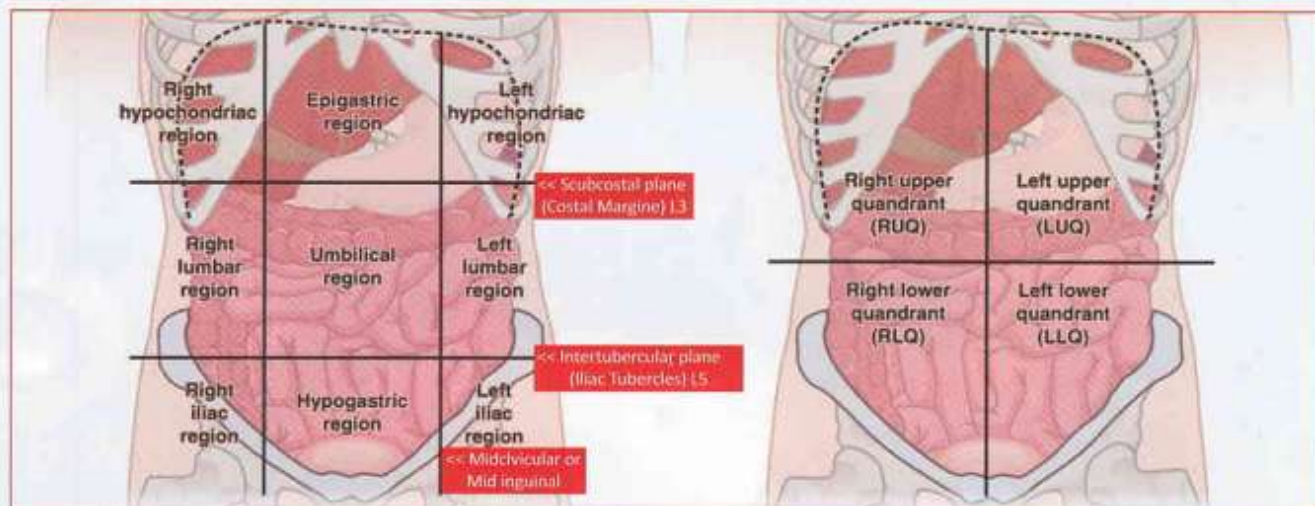
“

Rare presentations of common diseases are much more common than common presentations of rare diseases.

”

INTRODUCTION

► Surface anatomy of the abdomen:



The abdomen can be divided into quadrants by a vertical median plane and a horizontal transumbilical plane, which passes through the umbilicus.

- The liver and gallbladder are in the **right upper quadrant**.
- The stomach and spleen are in the **left upper quadrant**.
- The cecum and appendix are in the **right lower quadrant**.
- The end of the descending colon and sigmoid colon are in the left **lower quadrant**.

The abdomen also can be divided into nine regions by a midclavicular sagittal plane on each side and by the subcostal and intertubercular planes, which pass through the body transversely. These planes separate the abdomen into:

- three central regions (**epigastric, umbilical, hypogastric**)
- three regions on each side (**hypochondrium, lumbar, iliac**)

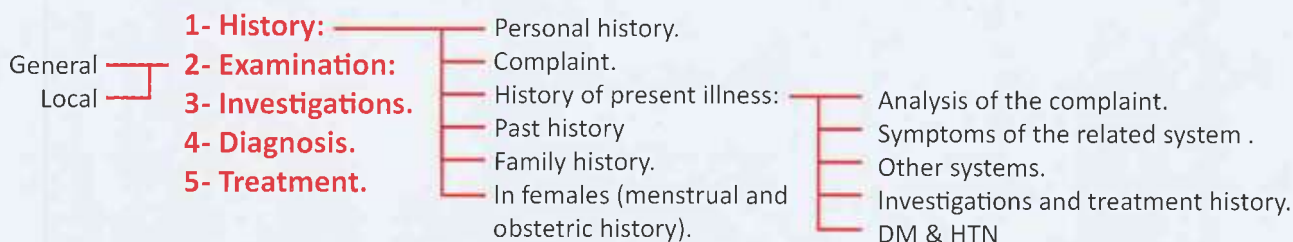
► N.B.

The level of the diaphragm varies during the breathing cycle. The dome of the diaphragm on the right can reach as high as the fourth costal cartilage during forced expiration.

Most of the liver is under the right dome of the diaphragm and is deep to the lower thoracic wall.

The inferior margin of the liver can be palpated descending below the right costal margin when a patient is asked to inhale deeply.

ABDOMINAL SHEET



ABDOMINAL HISTORY

► Personal History:

1. **Name:** اسم حضرتك ايه؟
to be familiar with the patient.
2. **Age:** عندك كام سنة؟
As certain diseases are more common in certain ages, e.g.
 - **Calcular obstructive jaundice:** middle aged females.
 - **Malignant obstructive jaundice:** old males.
 - **Cirrhosis:** usually in adults.
3. **Sex:**
 - **Gall Bladder stones:** females.
 - **Cancer head of pancreas:** male
4. **Occupation:** بتشتغل ايه؟
persons in certain occupations are more susceptible to certain diseases e.g.
 - **Farmers:** bilharziasis
5. **Marital state:** متزوج؟ عندك اولاد؟ كام؟ اصغرهم عنده كام سنة؟
for possible sterility or impotence.
6. **Residence:** ساكن فين؟
may reflect socioeconomic condition and may occasionally point to certain disease e.g.
Countryside: Bilharziasis
7. **Special habits:**
بتدخن؟ كام سيجارة في اليوم؟ بقالك كام سنة؟ بطلت في خلال هذه الفترة؟
بتشرب خمر أو مخدرات؟ بطريقة منتظمة ولا في المناسبات؟ نوع الخمر ايه؟
e.g. Alcoholism in cirrhosis.
8. **Mensterual History.**

► Complaint:

On patient's own words + duration؟ ايه اللي تايبك؟ ايه اللي جابك المستشفى؟ بقاله اد ايه التعب؟
(take the most recent and most important complaint and it should be brief)

Examples:

the patient is complaining of abdominal swelling of 3 week duration.
the patient is complaining of Vomiting of blood of 2 days duration.

► History of present illness (H.P.I.):

1. Analysis of complaint.
2. Symptoms of the related system.
3. Systemic Review:.
4. Investigation & treatment (related diseases).
5. DM & HTN

HPI should be:

- As long as possible and Contains medical terms.
- In chronological arrangement.
- In the form of a story.

1- Analysis Of The Complaint:

For analysis of any complaint in chest as usual (8) + 3 for any excreta.

8 as usual:

- Onset - course - duration. الشكوى ظهرت فجأة ولا بعد أد ايه؟ الأعراض ظهرت بالتدريج ولا على مدى أسابيع أو شهور؟ أو بتحدث في نوبات؟
- Association. هل كانت مصحوبة بحاجة وتذكر الأعراض الثانية اللي ممكن تكون موجودة؟
- What increase and what decrease. إيه اللي بيزود العَرَض ده؟ وإيه اللي بيقلله؟
- Effect of treatment. تأثير العلاج عليه؟؟ هل لما بتأخذ العلاج الأعراض بتقل ولا بتفضل زي ما هي؟
- Date of last attack. آخر مرة جالك العَرَض " وتقول إسم العرض ده " كان إمتى؟

+ 3 for any excreta:

- Amount.
- Content - Color - Consistency.
- Odour.

2- Symptoms of the related system: (3GPS)

- 3G** E
- **G.I.T. (upper & lower).**
 - **General (toxic).**
 - **Gynecological.**
- 3P** E
- **Pain.**
 - **Biliary system (hepatobiliary).**
 - **Urinary system.** مشاكل في البول
- 3S** E
- **Abdominal swelling.**
 - **Lower limb swelling.**
 - **Systemic review.**

1.GIT :

It's classified according to Ligament of treitz into upper & lower GIT symptoms:

Upper G.I.T.	Lower G.I.T.
1. Appetite. شهيتك على الأكل اتغيرت ؟؟؟ Anorexia / polyphagia. 2. Dysphagia. هل هناك صعوبة في البلع ؟؟ للسوائل أو للأكل ؟؟؟ 3. Vomiting. حصلك ترجيع ؟؟؟ 4. Haematemesis. رجعت دم قبل كده ؟؟؟ 5. Water brush. بتحس باللعاب عالي فمك ؟ ونفسك غمة عليك ؟؟ 6. Halitosis. رائحة فمك متغيرة ؟؟؟ 7. Heart burn: in GERD. هل عندك حمو على فم المعدة ؟؟؟ 8. Eructation. بتتكرع ؟؟؟ 9. Hiccough (idopathic / CRF). بيجينك زغطة كثير ؟؟؟	1. Flatulence. بطنك انتفخت ؟ وفيه غازات ؟؟ 2. Audible borborygmi. 3. Bowel habit. 4. Haematochezia. فيه دم أحمر مع التبرز ؟؟ 5. Melena. هل هناك براز أسود ؟؟؟

Appetite:**Polyphagia with weight loss:**

- Uncontrolled diabetes Mellitus.
- Thyrotoxicosis.
- Parasitic infestation.
- Malabsorption (Steatorrhea).

Anorexia:

Lack or loss of appetite for food

- Psychological (Anorexia Nervosa) in depression and other psychological troubles
- GIT: Gastritis
- Others: Cancers, TB, sever infection

Parorexia:

Disordered appetite, with craving for unusual foods

- Parasitic infestation with Ankylostoma
- Pregnancy
- Hypocalcemia
- Hysterical.

Dysphagia:

Dysphagia: is difficulty in swallowing & **Odynophagia** is Pain during swallowing.

Casues:

- **Mouth causes:** Glossitis & Stomatitis
- **Pharynx:** Pharyngitis, Tonsilitis & **Plummer Vinson Syndrome**
- **Oesophagus:** Foreign body, Corrosive aticture, **Achalasia & Carcinoma**

For more causes see the theoretical book...

	Achalasia	Carcinoma
Age:	Middle aged	Old
Sex:	Female	Male
Dysphagia for:	Fluids more	Solids more
Odynophagia:	maybe	no pain
General Condition:	good	very bad

Vomiting:

For analysis of any complaint in GIT as **usual (8) + 3 for any excreta (Stress on onset - timing - content)**

Onset:

- Spontaneous.
- Induced >> psychic

Timing:

Early morning	Late in the day	Relation to meal
Central causes	Pyloric obstruction	During >> esophageal. ½ hour after >> G. ulcer. 3 hours after >> D. ulcer.

Content

Mucous	Chronic gastritis - cancer
Bile	Intestinal obstruction
Fecal	Intestinal obstruction or fistula
Parasite	Ascaris
Stone	Fistula
Blood	See (Haematemesis)

Haematemesis:

For analysis of any complaint in chest as **usual (8) + 3 for any excreta.**

Preceded by **Vomiting** and followed by **Melena Dark, tarry stool**

DD:

	Haemoptysis	Haematemesis
Preceded by	Cough	Nausea & vomiting
During		
• Contents	Froth	Food
• Color	Bright red	Dark
• PH	Alkaline.	Acidic
Followed by	Blood tinged sputum	Melena.

Halitosis:**D.D. of Halitosis:**

1. Foetor hepaticus: liver cell failure.
2. Amoniacal: renal failure.
3. Aceton: D.K.A.
4. Foetoid: S.L.S.
5. Local: dental caries.
6. Dietary.
7. Achalasia of cardia

Hiccough:

Definition: abnormal spasmodic contractions of the diaphragm (phrenic nerve irritation).

Aetiology:

General	Chest	Neck	Abdomen
• CRF • Psychic	• Carcinoma • Empyema • Pericarditis	• Rapid thyroid enlargement	• Subphrenic abscess • Hepatic abscess • Splenic abscess • Acute gastric distention • Haemo peritoneum

Changes in the bowel habit:**Diarrhea:**

A condition of having at least 3 loose or liquid bowel motions per day, provided that patient on western diet ;
or: Passage of > 250 gm stool per day

Mechanisms of diarrhea:

- **Hyper motility:** Exaggerated gastro-colic reflex e.g. D.M. - thyrotoxicosis.
- **Osmotic:** Related to food e.g. lactulase.
- **Secretory:** Infection - U.C.
- **Spurious:** after prolonged constipation due to fermentation - rectal masses.
- **Steatorrhea.**

For more causes see the theoretical book...

Constipation:

Presence of two or more of the following:

- Infrequent stools < 3 per week
- Hard stools (> third of motions are hard)
- Incomplete evacuation (> third of motions)
- Straining (> third of motions)
- Manual evacuation

For more causes see the theoretical book...

Tensmus:

Desire of defecation without an actual motion

Dysentery: Bloody stool (Diarrhea + tenesmus)

(passage of mucus & blood with stool) هل فيه ثقل أو تعنية

مع البراز ???

Causes

- **Infective:**
 - Bacterial: bacillary.
 - Parasitic: amoebic and B.
- **Metabolic:** C.R.F.
- **Toxic:** Mercury poisoning.
- **Local causes:** diverticulosis, ulcerative colitis "U.C.", cancer colon or rectum.

For more causes see the theoretical book...

Melena:

Passage of soft black tarry offensive stools, due to upper GIT bleeding (from the oropharynx to end of mid gut) هل هناك براز أسود زي الزيت

Site: above ileo-cecal junction.

Amount: not less than 60 ml.

Duration: > 8 hours.

If less than 60 ml, it is called occult blood in the stool.

If the level below the ileo-cecal junction or the duration less than 8 hours, it is called bleeding per rectum.

D.D.:

- Iron therapy or Bismuth:
- History of drug intake & constipation.
- Hard stool with non offensive odour.

For more causes see the theoretical book...

2. Genral :

Toxic manifestations: FAHM (fever, headache, anorexia and Malaise):

in patients with Hepatitis, L.C.F., Hepatocellular carcinoma & TB enteritis ... etc

3. Gynecological symptoms in female :

Bleeding, Discharge, mass, Hirsutism & hot flushes

4. Pain :

Don't forget (11 points) : 8 as usual + 3 (site, radiation and character).

Types of Abdominal Pain:

Visceral	Parietal (Somatic)	Referred
• Felt at the site of primary stimulus • vague & Diffuse • Types: • Colic: from hollow organs (Obstruction) • Dull ache: from hollow or solid organ (Distention)	• Irritation of the parietal peritoneum • Sharp & Localized	• pain felt at a site other than the site of the stimulus but in the area supplied by the same neural segment • e.g. cholecystitis

Medical causes of Acute Abdominal Pain:

- Addisonian Crisis
- Porphyria
- severe hyper calcemia
- Hemolytic Anemia
- Sickle cell anemia
- Myocardial Infarction
- Acute pleurisy & pneumonia
- Diabetic Ketoacidosis
- Systemic lupus erythematosus
- Rheumatic fever (non infective peritonitis)
- Familial mediterranean fever
- Food poisoning & uremia

Extra-Abdominal Causes of Abdominal Pain:

- Myocardial Infarction
- Herpes zoster
- Acute Pleurisy & Pneumonia
- Dissecting aortic aneurysm
- Torsion of testis & ovary
- Cord compression (Thoracic segment)

► N.B.

Recurrent abdominal pain

P.U. - Peritonitis - Pancreatitis - chronic diverticulosis - cancer colon - chronic I. O.

Types of splenic pain:

Stretching of the capsule	Stretching of ligament	Splenic infarction	Splenic abscess
Dull aching	Dragging	Perisplenitis Stitching \ stabbing	Throbbing

5. Biliary (Hepatobiliary) :

Symptoms:

- Jaundice.
- Pruritis.
- Dyspepsia.
- Pain.

► N.B.

Jaundice may be:

- Viral : acute - short - regressive.
- Calcular : acute intermittent.
- Malignancy : gradual - progressive (2 Months).
- Chronic liver disease : gradual - intermittent (2Y).

► Jaundice :

Definition:

yellow discoloration of skin & mucous membranes due to increased level of bilirubin > 2.5 mg %.

Analysis of the jaundice:

1. Onset:

- **Acute:** viral hepatitis, calcular obstruction.
- **Gradual:** malignant obstruction, cirrhosis.

2. Course:

- **Progressive:** malignant obstruction, cirrhosis.
- **Regressive:** viral hepatitis.
- **Intermittent:** calcular obstructive jaundice, periampullary carcinoma, hemolytic jaundice, chronic active hepatitis.

3. Duration:

- **Short:** viral hepatitis.
- **Long:** cirrhosis.
- **More than 2 years exclude malignancy.**

4. Urine:

- **Dark:** hepatocellular & obstructive.
- **Pale:** hemolytic.

5. Stool:

- **Pale clay:** in obstructive.
- **Dark:** in hemolytic.
- **Slightly pale** in hepatocellular.

6. Anorexia, nausea, vomiting:

Occur at the onset of viral hepatitis.

7. Fever:

- **Hepatocellular:** pre-icteric phase of viral hepatitis.
- **Hemolytic:** during hemolytic crisis - malaria - incompatible blood transfusion.
- **Obstructive:** Charcot's triad.

8. Bleeding tendency: from skin, orifices in:

- Liver cell failure.
- Obstructive jaundice.

9. Pain:

- **Hepatocellular:** dull-aching pain in right hypochondrium in case of viral hepatitis.
- **Hemolytic:** bone pain and abdominal pain in hemolytic crisis.
- **Obstructive:**
 - Biliary colic (calcular obstruction).
 - Epigastric pain radiating to the back (malignant obstruction).

10. Pruritis:

- In obstructive jaundice.
- Primary biliary colic.

6. Urogenital System

Micturation:

- Pain.
- Micturation difficulties (dysuria).
- Urine abnormalities: volume - colour - frothy.
- Uraemic symptoms.

Uraemic Manifestations:

- Anorexia
- Halitosis (Bad oral odor)
- Hicough
- Disturbed consantration

Sexual:

- Desire.
- Erection troubles.

7. Abdominal Swelling :

Localized	Generalized
• Spleen. • Liver. • Mass.	• Ascites >> History of aspiration: Amount, Aspect, Colour (red = hge.), Complications: fever (infection), wound sepsis, shock & hepatic encephalopathy.

8. Lower limb Swelling :

- L.C.F.
- Bilharziasial Corpulmonale.

► N.B.

Causes of Lower limb edema in LCF are:

1. Hypoalbumenemia
2. Salt and water retention

► N.B.

Causes of ascitis in LCF are:

1. Hypoalbumenemia
2. Salt and water retention
3. portal HTN as a localizing factor
4. Lymphorrhea
5. Associated conditons as SBP or malignancy

9. Systemic review :

3. Other Systems:

CVS: Systemic congestion, pulmonary congestion & low cardiac output manifestation as syncope

Neurology: motor and sensory affection

if nothing involved write

eg. no symptoms suggest other systems affection

4. Investigation & treatment (related diseases):

eg. patient investigated by LFT & US

► Past History :

Diseases	Operations	Drugs
• Bilharziasis. • Hepatitis. • T.B. • D.M. • Hypertension.	History of blood transfusion.	Hepato-toxic • Dose dependant Paracetamol 16 gm. • Non dose dependant • Acute necrosis : INH - PAS. • Cholestasis : aldomet. • Tumors : C. pills

Bilharziasis	hepatitis	
	Subclinical	Clinical
History of contantct to canals. Clinical picture of Bilharziasis. Investigations	Unnoticed	Fever. Jaundice. Dark urine. Fever hospital.
Easily excluded	Can't excluded	

► N.B.

Bilharziasis **NEVER** cause Liver Cell Failure

**G.I.T. (upper)**

شهيته على الأكل تغيرت ؟؟ هل هناك صعوبة في البلع للسوائل أو للأكل ؟؟
 حصىك ترجيع ؟؟ رجعت دم قبل كده ؟؟
 يتحسن باللعب مالي فمك ؟؟ ونفسك غامة عليك ؟؟
 رائحة فمك متغيرة ؟؟ هل عندك حمو على فم المعدة ؟؟
 بتتكرع ؟؟ بجيالك زغطة ؟؟

G.I.T. (lower)

Flatulence بطنك انتفخت ؟؟ وفيه غازات ؟؟
 haematochezia فيه دم أحمر مع التبرز ؟؟
 melena هل هناك براز أسود زي الزيت ؟؟
 فيه إسهال أو تعنية ؟؟ فيه أصوات من البطن ؟؟

General

فيه ارتفاع في درجة الحرارة ؟؟ فيه صداع ؟؟
 فيه فقد للشهية ؟؟
 فيه توعك وهمدان في جسمك ؟؟

Gyneacological

فيه ألم ؟؟ فيه نزيف ؟؟ فيه إفرازات ؟؟

Pain

عندك ألم في بطنك ؟؟ بيزيد بإيه ؟؟ ويقل بإيه ؟؟ ومتناسخ ال 11 نقطة

Biliary system

لون عينيك بقى أصفر ؟؟ ولون البول والبراز اتغير ؟؟

Urinary and genital system

مشاكل في البول
 فيه ألم عند التبول ؟؟ البول بيسببك ؟؟
 عندك ضعف جنسي ؟؟

Abdominal Swelling

بطنك ورمت ؟؟ أو انتفخت ولا لا ؟؟

Lower limb Swelling

رجلك ورمت أو تُلُخِيت ولا لا ؟؟

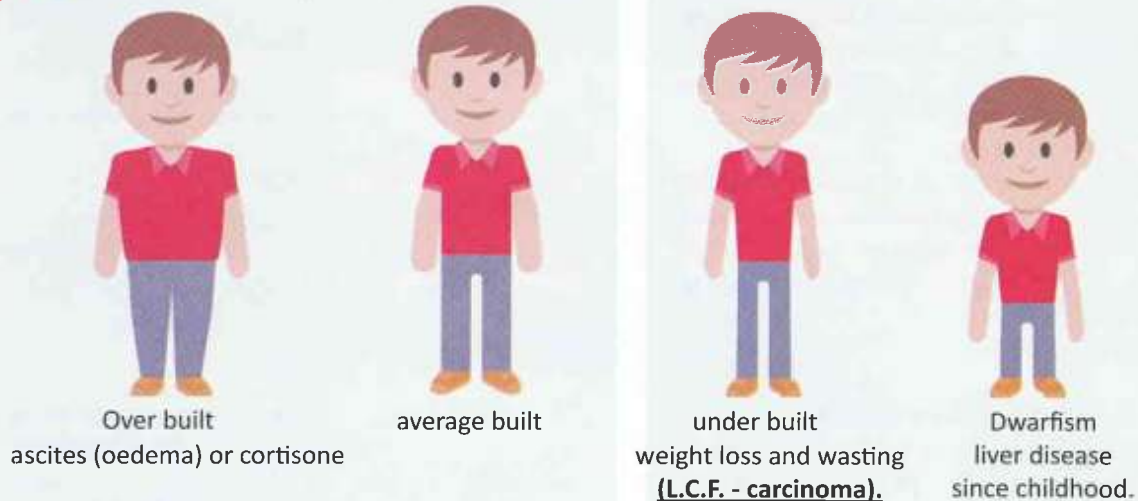
Systemic review

فيه حاجة تانية بتشتكي منها في بطنك ؟؟
 وتسال المريض برده
 : وتقوله
 هل عندك مشاكل في أي جزء ثاني في جسمك ؟؟
 عملت فحوصات إيه ؟؟
 وأخذت علاج إيه ؟؟

GENERAL EXAMINATION

A أرقام **Vital Signs: Blood pressure, pulse, Respiratory rate, Temperature.**

- **Tachypnea and Respiratory distress:** tense ascites - pleural effusion.
- **Hypotension:** shocked - postural hypotension (VDMs).
- **Pulse:**
 - **Tachycardia:** hyperdynamic circulation (VDMs).
 - **Bradycardia:** in obstructive jaundice.
- **Fever:** low grade.

B **Built:** Over built, average built or under built**C** **Complexion:**

- Pallor:** Anemia (all types may occur).
Jaundice (types).
Hyperpigmented: (Hemochromatosis - L.C.F.).
Cyanosis: tense ascites - porto-pulmonary - A.V. shunt.

**D** **Decubitus or position**

- Orthopnea:** tense ascites - pleural effusion.
Platypnea: hepatopulmonary syndrome: see below

E **(H & N): Congested neck vein: tense ascites - pleural or pericardial effusion.**

upper limb

lower limb: Lower limb edema

Flapping tremors in Hepatic pre-coma

Clubbing

- Crohn's - Ulcerative colitis.
- Biliary cirrhosis.
- B. polyposis.
- Intestinal steatorrhea.

F فكر **Mentality:** Disturbed mentality (**Hepatic encephalopathy**).**Face** Temporalis muscle wasting + enlarged parotid.

► Functional evaluation of patient with liver cirrhosis :

Patient with cirrhosis may be compensated or decompensated (vascular or parenchymatous)

► Vascular decompensation (portal hypertension) :

Evidences of portal hypertension:

- **Splenomegaly:** the single most important diagnostic sign. And may lead to hypersplenism.
- **Ascites:**
 - **Early ascites:** acute portal or splenic vein thrombosis.
 - **Late:** related to additional factors as hypo-albuminaemia.
- **Portosystemic anastomosis:**

Location	Portal circulation	Systemic circulation	Consequences
Gastro-esophageal	Coronary V. of stomach	Azygos vein	Gastro -esophageal varices
Anorectal	Middle and superior hemorrhoidal V.	Inferior hemorrhoidal vein	May be mistaken for hemorrhoids
Anterior abdominal wall	Para-umbilical V. in falciform lig. (left portal vein)	Superior & inferior epigastric V.	Caput medusae
Retroperitoneal	• Splenic V. branch. • Veins around liver and diaphragm. • Visceral veins.	• Left renal V. • Abdominal wall V. • Retroperitoneal V.	Between: • Liver & diaph. • Spleen & diaph. • Intestine & post abd. Wall.

Presentation of Portosystemic anastomosis:

1. Asymptomatic (diagnosed by ultrasound or endoscopy).
2. Haematemesis.
3. Internal bleeding.
4. Chronic encephalopathy.

► Parenchymatous decompensation (L.C.F.) :

Manifestations of liver cell failure:

- **Jaundice D.D.**
- **Encephalopathy.**
- **Ascites.**
- **Fever (low grade).**
- **Fetor hepaticus D.D.**
- **Flapping tremors.**
- **Fatigue.**
- **Skin changes:**
 - Spider naevi D.D. - palmar erythema - white nails (Decreased protein).
- **Endocrinal**
 - Gynecomastia.
 - Testicular atrophy.
 - Feminine pubic hair.
 - Hyperaldosteronism.
- **Blood**
 - Anaemia (pallor).
 - Bleeding tendency.
- **C.V.S:**
 - Hyperdynamic circulation.
- **Kidney:**
 - Hepato-renal failure.

For more details about liver cell failure see the theoretical book ...

► Spider Naevi :

Central dilated arteriole with radiating capillaries (looks like a spider).

Pressure on central arteriole leads to fading.

Releasing of pressure: the blood refill the "legs of the spider".

Distributed in the upper half of the body (course of the S.V.C.).

Confirmed by:

- Central pressure = blanching.
- Slide test = pulsation.

One or two spider naevi may occur in normal people, thyrotoxicosis or pregnancy.

More than 5 considered as significant and likely to due to L.C.F.



DD:

	Level	Pruritus	Pressure	Site
Spider naevi	Raised	Not	Fading	SVC course
Insect bite	Raised	Itchy	No	Exposed area
Purpura	Not	Not	Not	LL (common)
Campbell de Morgan spots	Raised	Not	Not	Trunk
Venous star	variable	±	Fading	LL

► Palmar Erythema :

redness of thenar, hypothenar, distal ends of metacarpal bone, pulp of fingers with central pallor.

Causes:

- Pregenancy.
- Thyrotoxicosis.
- L.C.F.
- Estrogen containing contraceptive pills.
- Familial

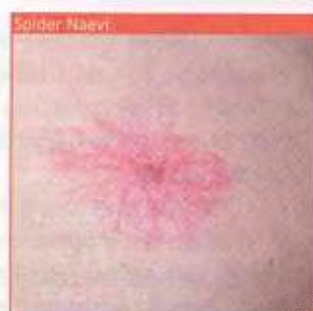


► Additional Items of General Examination in patient with liver disease :

1. Limbic examination.
2. Head and neck examination.
3. Sexual chr. Evaluation.
4. Systemic review (C.N.S. - Chest - C.V.S. - Kidney).

1. Limbic examination:

Upper limb	
Nails	• Koilonychia (Fe deficiency due to GIT bleeding). • Leuconychia (hypoalbuminaemia). • Muehrke's lines (hypoalbuminaemia). • Blue lunulae (Wilson's).
Hands	• Asterixis: patient stretches out hands in policeman's stop position and fingers spread out (+ closed eye). • Palmer erythema. • Pallor of the creases. • Dupuytren's contracture: (fibrosis of the palm's fascia) (alcoholism, manual labor). • Palmer or tendon xanthomata.
Arms	• Scratch marks. • Spider naevi. • Bruising. • Xanthomata.
Lower Limb	
	• Oedema. • Bruising. • Neurological examination. • Xanthomata.



► N.B.

Enlarged lymph node due to:

Supraclavicular nodes (Virchow's node): lung or GIT malignancy.

2. Head and neck :

Eyes	• Cornea rings (Wilson's). • Sclera: jaundice. • Iritis: inflammatory bowel disease. • Xanthelesma.
Lips	• Telangiectasia. • Brown freckles (Peutz-Jaeghers).
Mouth	• Breath: fetor hepaticus - Ethanol (alcoholism). • Ulcer: (Crohn's, coeliac) - white candida patches - cracks at mouth edges. • Teeth: cavities (acid effect). • Gums: hypertrophy - bleeding - Gingivitis.
Tongue	• Leucoplakia: smoke, spirits, sepsis, syphilis, sore teeth. • Atrophic glossitis: hypovitaminosis - Plummer vinson. • Macroglossia : B12 deficiency.

**3. Sexual character evaluation:**

e.g. gynecomastia and testicular atrophy.

4. Systemic review:

	Cause	Result	Association
C.N.S.	Autoimmune PN (gastro paresis diabeticorum)	Hepatic encephalopathy	Hepatolenticular "Wilson's"
Chest	Cor-pulmonale → enlarged tender liver.	Amoebic liver abscess. Hydatid. Sympathetic effusion (Rt. PI). Hepato-pulmonary Syndrmome.	Alpha one anti trypsin deficiency
C.V.S.	Cardiac cirrhosis. S.B.E. = splenomegaly.	Bilharziasis. Cor-pulmonale.	
Renal		Hepato-renal. Pre-renal (acute GIT bleeding). Nephrotic (extra hepatic of HCV - B man-soni).	

► N.B.**Hepatopulmonary syndrome:**

In advanced LCF → Marked ↑ vaso dilator materials (VDM) → opening of pulmonary A-V shunt → Cyanosis, Hypoxia & Dyspnea.

On laying flat

→ ↑ venous return → shunt become inactive, so dyspnea improves but appears on standing (Platypnea)

LOCAL ABDOMINAL EXAMINATION

Inspection

General consideration:

- The patient must be exposed from the nipple to the symphysis pubis.
- The patient's hand should remain at his sides with head resting on a pillow.
- Flexion on the knees may relax the abdomen.
- The patient should have an empty bladder.
- Warm room and adequate light must be provided.
- For easy localization of any abnormalities it is useful to divide the abdomen in 4 quadrants or 9 segments
- Watch the patient's face for signs of discomfort during the examination.

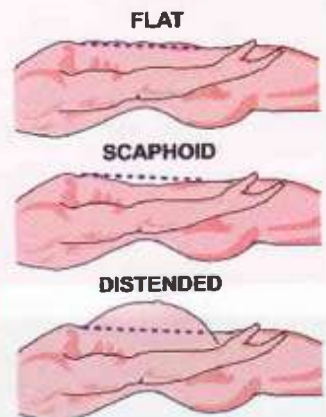
Inspection and confirmed by Some Palpation

Front (15) = Contour+7+7		Back (5)
Midline (7)	Sides (7)	
1. Subcostal angle. 2. Epigastric pulsation. 3. Visible peristalsis. 4. Divercation. 5. Umbilicus. 6. Hair distribution. 7. Genitalia.	1. Respiratory movement. 2. Breast. 3. Hernial orifices. 4. Pigmentation. 5. Veins. 6. Scratching marks. 7. Scars.	1. Scar. 2. Deformity. 3. Swelling of renal angle. 4. Cafe' au lait patches "neurofibroma". 5. Tuft of hair "spina bifida".

► Contour :

(must be the first step) can be best appreciated by standing at the foot of the table and looking up towards the patient's head.

- **Normal:** slightly scaphoid with preserved waist.
- **Abnormal:**
 - **Scaphoid:** the anterior abdominal wall is sunken and presents a concave in Cachectic patient.
 - **Bulging:**



Symmetrical (5 Fs)	Asymmetrical
• Fat = obesity (sunken umbilicus) • Flatus = (gases) • Foetus • Fluid = ascites • Full urinary bladder	• Organ swelling (HSM) • Ovarian or uterine tumor • (comment on mass)

Inspecting the abdomen (abdominal contour)



Distended abdomen with full flanks



Scaphoid



Inspection of the front (midline) (7) :**1. Subcostal angle**

- **Normal:** acute to right (70° - 90°).
- **Obtuse:** upper abdominal swelling (HSM) - ascites - barrel shaped chest.

2. Epigastric pulsation:

Place your hand longitudinal in the subcostal angle, Pulsation classified according to the direction into :

- **At the tip of fingers:** RV ++ , increased with deep inspiration.
- **From the right side:** **hepatic pulsation** (tricuspid stenosis - TR), liver is enlarged and tender during bimanual examination.
- **From behind:** **aortic pulsation** (thin - hyperdynamic - aneurysm), pulsating down to umbilicus.

3. Visible peristalsis.

- **Normal:** in thin person or in emaciated person.
- **Abnormal:**
 - **Pyloric obstruction:** slow waves from the left rib margin to the right. Exaggerated by massage, tapping or drinking soda + confirmed by Succussion splash (see auscultation).
 - **Intestinal obstruction:** called step ladder.

4. Divercation of recti :

- Separation of rectus abdominis muscles
- Ask the patient to raise his head up without support (lift head).
- Usually confirmed by palpating the defect between both recti.
- **Causes:**
 - chronic increase of the intra abdominal pressure (HSM - ascites) + hypoproteinemia.

5. Umbilicus (4 Ss + CD)

Site	• Normal: midway between symphysis pubis and xiphisternum. • Upper abdominal swelling and ascites slightly shift it downward, vise versa.
Shape	• Normal: Inverted. • Everted umbilicus means chronic increase of the intra-abdominal pressure (HSM – ascites).
Swelling	• Hernia: expansile impulses with cough. • Nodules: GIT of breast cancers (Sister Mary <i>Joseph's nodule</i>).
Skin lesions	• Scar. • Wound.
Colours	• Cullen's sign: intra-peritoneal bleeding - acute pancreatitis " • Causes of generalized pigmentation (see general ex VI).
Discharge	• Urine: patent urachus. • Faeces: colonic fistula. • Pus: Suppurative inflammation. • Bile: biliary fistula. • Fluid: ascities (bursting abdomen).

Examining Subcostal Angel



Palpating epigastric Pulsations



Checking Recti Divercation



Checking for impulse on cough

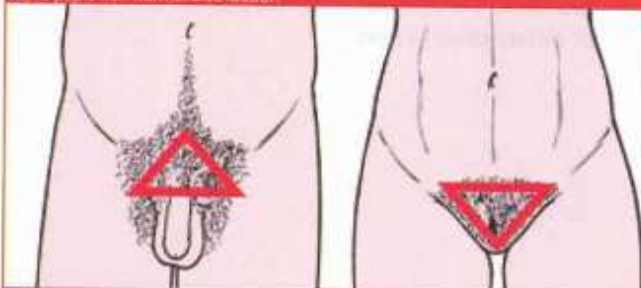
**6. Supra pubic hair distribution:**

- **Normal:** triangular in male with apex directed upward (umbilicus) & horizontal in females.
- **Abnormal:** as in
 - **L.C.F.:** feminine distribution.
 - **Hypogonadism:** loss of hair.

7. External genitalia

If the patient is circumcised - phimosis - para phimosis etc

Supra pubic hair normal distribution



► Inspection of the front (sides) (7) :

1. **Respiratory movement (see lung disease)**

- Patient of tense ascites usually orthopnic and breath thoraco-abdominal.
- Patient of peritonitis usually has rigid dorsal decubitus and has absolutely thoracic breathing.

2. **Breast:** (one of the general examination items and must be examined in local abdominal examination).

- **Gynaecomastia:** hyperplasia of the glandular component of male breast. Confirmed by pinching test as a disc (button like). Usually presented as a one sign of L.C.F. or caused by spironolactone. Comment on: laterality - tender or not.
- **Breast atrophy** in females noticed in L.C.F.

3. **Hernial orifices:**

- Expansile impulses with cough.
- Preferred in standing patient.
- **Palpate the orifices defect:**
 1. **Epigastric:** small, midline through linea alba defect located between the xiphoid process and umbilicus.
 2. **Umbilical:** bulging defect at umbilicus.
 3. **Incisional:** defect in abdomen muscles after surgical incision. Must palpate the size of the defect.
 4. **Abdominal:** hernia through the abdominal wall.
 5. **Inguinal:** Direct or Indirect.

4. **Veins**Visible or dilated:

- **Visible:** straight, narrow and not raised.
- **Dilated:** Tortuous, wide and raised above the level of the skin.

If dilated:

Prominent, dilated veins may represent collateral circulation through the abdominal wall that has developed to compensate for obstruction of either the inferior vena cava or portal vein.



Portal (caput medusa)	Systemic (IVC obstruction)
Central	Peripheral
By Milking test: Filling away from umbilicus	By Milking test: From down to up

5. **Pigmentations:**

- Bluish at umbilicus: Cullen's sign (bleeding in peritoneum).
- Bruises on flanks: Grey Turner's sign (retroperitoneal bleeding as pancreatitis).
- Jaundice: yellow skin – usually due to liver disease or biliary obstruction.



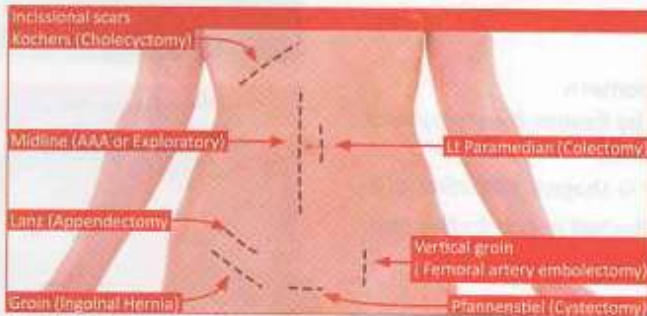
6. Scratch marking:

Denotes pruritus usually with obstructive jaundice

- Multiple, parallel and superficial.
- In accessible area.

7. Scars:

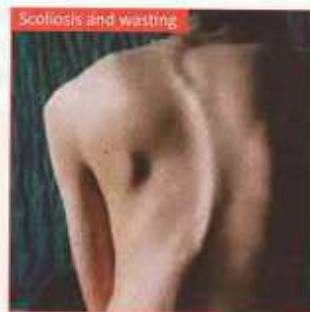
- Surgical (name - healing).
- Traumatic.
- Cautery.

**► N.B.****Add for inspection in all views**

- Scar:
- Surgical (name - healing).
- Traumatic.
- Cautery.
- Striae.
- Scratch marking.
- S.C. bleeding.
- Pigmentation.
- Elasticity (dehydration).
- Oedema.

inspection of the back (5) :

1. Scar.
2. Deformity.
3. Swelling of renal angle. (see later)
4. Cafe' au lait patches (neurofibroma).
5. Tuft of hair "spina bifida".

**Palpation****General rules:**

- Let you patient lay on bed comfortably.
- Ask him or her to expose from above nipple to mid thigh (it's accepted to cover genitalia and breasts (in females) until it's needed to be examined).
- Ask the patient to bend his knee.
- Warm your hands.
- Ask patient if any part is tender: examine that last.
- Abdominal muscles must be relaxed by patient knee flexion.
- Superficial palpation then deep palpation.

Normal palpable structures:

- Sigmoid colon: left lower quadrant - firm, narrow tube.
- Cecum and ascending colon: right lower quadrant - a softer, wider tube.
- Pulsation's of ascending aorta: midline in upper abdomen.

Types of Palpation:

- **Superficial palpation**
- **Deep Palpation which have many methods:**
 - Ordinary method
 - Enforced
 - Bimanual
 - Ballotment
 - Dipping
 - Rolling
- **Per Rectal (PR)**
- **Per Vaginal (PV)**

Less commonly palpable, but normal:

- Liver: just below right costal margin (soft).
- Transverse and descending colon.
- Lower pole of the right kidney: (very deep, mostly in thin women).
- Iliac artery: pulsation's - left lower quadrant and right lower quadrant.

► **Superficial palpation :****Aim:**

- Gain patient's confidence
- Detect tenderness, hyperesthesia, temperature, guarding & rigidity.
- Detect superficial swellings e.g. lipoma, varicosities & hernia:
Ask the patient to contract the abdominal wall muscles by raising the head (differentiate between intra & extra-abdominal swelling and notice the swelling mobility with respiration).
- To prepare the patient for deep palpation

Method:

- Place your extending hand flat on the patient's abdomen
- Gently press (1CM depth) the patient's abdomen by flexion & extension of the metacarpal joints of fingers (light dipping method)
- slowly progress around the 9 regions either in S or G shaped palpation & if the patient complains of abdominal pain let the affected region to the end

Finding:**Assessment of muscle tone:**

There are 3 reactions that indicate pathology:

- **Guarding:** muscles contract as pressure is applied.
- **Rigidity:** rigid abdominal wall indicates peritoneal inflammation.
- **Rebound tenderness:** release of pressure causes pain.

	Guarding	Rigidity
Mechanism	Voluntary contraction of anterior abdominal wall	involuntary contraction of anterior abdominal wall
Respiration	Disappears on inspiration	present with inspiration
Movement with respiration	Freely mobile	Limited
e.g.	During painful palpation	Peritonitis

**Examination of patient with guarding (Nickson's technique):**

Make a fist by your lt. hand and press by it on the sternum, while examining abdomen by rt. hand; this will limit the chest muscles of respiration allowing the abdominal muscles to move which will release the guarding.

► N.B.

If the patient is frightened, initially use the patient's hand under yours as you palpate. When patient calms then use your hands to palpate. The most sensitive indicator of tenderness is the patient's **facial expression**.

► **Deep palpation :****Aim:**

- Localize abdominal organs
- localize abdominal masses

Method:

- Each organ has a special method for palpation, see below...

Finding:

- Describing the mass as the following:
 - edge
 - consistency
 - moves with respiration or not
 - tenderness
 - pulsations
- Intral or extra abdominal
- site
- size
- shape

Palpation of Liver

Surface anatomy of the Liver:

The liver can essentially be visualised as a triangle, with its upper margin below the nipples on either side of the chest, and the lower margin making a line from just above the tenth rib on the right side to below the nipple on the left side.

The superior surface of the liver lies just below the diaphragm; this means that the lower margin of the liver will move downwards on inspiration, and this can be palpated. As the liver is also a very dense organ, it is very dull to percussion and you can easily percuss out the borders of the liver if palpation is unsuccessful. The gallbladder area can be palpated around the tip of the right ninth rib. The normal gallbladder is impalpable; it only becomes palpable when distended with stones or bile, and the area will become very tender if there is inflammation present.

Liver Span

- liver span of Rt lobe around 8: 12 cm in adult.
- Lt lobe span 4: 8 cm in adult but its ultrasonographic finding

► Liver Palpation :

- Mark the upper border 1st by tidal percussion .
- The lower border by:
 1. Ordinary technique.
 2. Bimanual.
 3. Hooking.
 4. Dipping.
 5. Tip of hands (Hutchinson's method).
 6. Auscultatory method (Macleod).

1. Ordinary technique:

- **Right lobe** at MCL beginning from right iliac region.
 - **Left lobe** at midline from umbilical region.
- Ask the patient to take a deep breath. By radial aspect of the index or tip of fingers, Feel the edge of the liver press against your fingers. Or, it may slide under your hand as the patient exhales.

2. Bimanual:

As ordinary with the **left hand** placed just below the right costal margin.

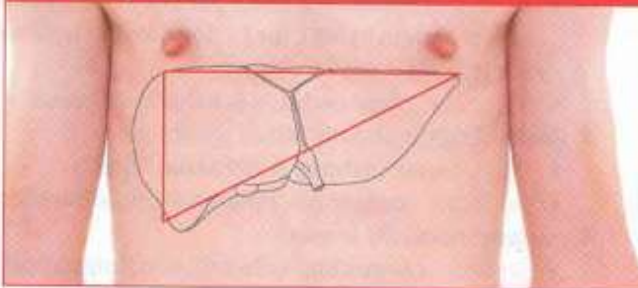
3. Hooking method:

- Stand by the patient's chest.
- (Hook) your fingers just below the costal margin and press firmly.
- Ask the patient to take a deep breath.
- You may feel the edge of the liver press against your fingers.

4. **Dipping:** (for palpation of organs in tense ascites)
Flex MCP joint fast to displace fluid and palpate a mass or enforced by both hands.

Disadvantage: The organ descriptive details are impossible

surface anatomy of the liver:



Borders:

Upper border	Lower border
• Lt - MCL 5th intercostal space. • Rt - MCL 5th rib. • Rt - MAL 7th rib. • Rt - scapular 9th rib.	• Lt - MCL 5th intercostal space. • Lt 8th costal cartilage. • Midway between the xiphisternum & umbilicus. • Rh 9th costal cartilage. • Rt - MCL 1 inch below the costal margin. • Rt - MAL 11th rib.

Bimanual Liver Palpation



Ordinary Liver Palpation



Hooking method



Ordinary method



Enforced method



Dipping method



Comment on following items:

- Site:** in right hypochondrium
- Size:** normally not felt below costal margin
 - Enlarged:** measure it using patient fingers below costal margin in MCL for Rt. lobe and in midline for Lt. lobe
 - Shrunk:** liver cirrhosis & Bilharzial fibrosis
- Border Edge:** normally felt as resistance
 - Sharp:** liver cirrhosis & Bilharzial fibrosis
 - Rounded:** congenital, inflammatory, infiltration
- Surface:** normally smooth
 - Smooth:** congestion, inflammation, infiltration, cirrhosis
 - Irregular:** liver cirrhosis if macro-nodular
 - Nodular:** malignancy
- Consistency:** normally soft
 - Soft & enlarged:** congenital, inflammation & infiltration
 - Firm:** Bilharziasis
 - Hard:** malignancy
 - Cystic:** amebic abscess & hydatid cyst
- Tenderness:** Test to examine tenderness: fist test in non palpable live Congestion, inflammation & malignancy
- Pulsation:** TR, TS, hemangioma
 - Examination for pulsation by bimanual examination: put one hand on the liver anteriorly and the other hand at the back, ask the patient to hold his breath and feel for pulsation.

► Example:

Intraabdominal swelling, moves up & down with respiration, in Rt. hypochondrium, 3 fingers below costal margin, not tender, not pulsating, sharp edge, firm in consistency, smooth surface, mostly it's Rt lobe of liver

► Causes of hepatomegaly :**Infectious:**

- Leptospirosis
- Pyogenic cholangitis
- Pyelophlebitis
- Portal pyemia
- Amebic hepatitis
- Toxoplasmosis
- Viral hepatitis
- Herpes simplex

Metabolic:

- Wilson's disease
- Hemochromatosis
- Fatty liver e.g. Reye's, DM
- Lipid storage disease
- Glycogen storage disease
- (Von Geirkes)
- Blood:**
- Megaloblastic anemia

Circulatory:

- Chronic venous congestion
- Congestive heart failure
- Pericardial effusion
- Constrictive pericarditis
- Conginital:**
- Riedel's lobe
- Polycystic disease

Tumors:

- CMV
- Metastasis
- Hepatoma
- Cholangioma

Palpation of Spleen**Surface anatomy of the spleen:**

Site: left hypochondrium

Under 9th, 10th & 11th ribs (long axis on 10th rib).

Medially: scapular line.

Laterally: MAL.

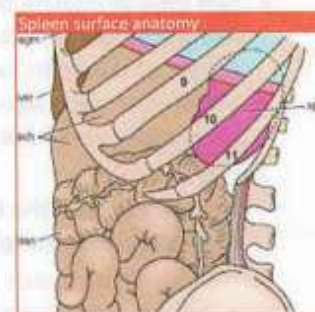
► Spleen Palpation :

- Ordinary technique.
- Bimanual.
- Hooking.
- Dipping.
- Tip of hands (Hutchinson's method).

1. Ordinary technique:

Place your right hand flat on the right iliac fossa and direct your hand towards the left costal margin & feel the edge of enlarged spleen during the peak of inspiration (or ask the patient to take deep breath in) in one of the following position:

- Radial side of index finger: parallel to the left costal margin
- Tips of fingers (middle three fingertips): perpendicular to the left costal margin



2. Bimanual:

As ordinary with the **left hand placed just below the Left costal margin**. If not felt do the same method in Rt. lateral position

3. Hooking method:

- Stand by the patient's chest on the **Lt. side**.
- (Hook) your fingers just below the costal margin and press firmly.
- Ask the patient to take a deep breath.
- You may feel the edge of the spleen press against your fingers.

4. Dipping: (for palpation of organs in tense ascites)
Flex MCP joint fast to displace fluid and palpate a mass or enforced by both hands.

Disadvantage: The organ descriptive details are impossible

**Comment on the following items:**

- 1. Site:** in left hypochondrium
- 2. Size:** normally Not felt below costal margin : normally or impalpable spleen
 - **Palpated:** enlarged at least 3 times if enlarged & doesn't cross the midline (or umbilicus): count by patient's finger breadth
 - **crossed midline** : huge spleen
- 3. Border Edge:**
 - in bilharziasis has a rounded anterior edge with one or multiple notch
- 4. Surface:** in bilharziasis smooth
- 5. Consistency:**
 - **Soft & enlarged** : in endocarditis, Malaria & septicemia

6. Tenderness:

In inflammation : typhoid & Brucellosis, TB (milliary), SBE, infarction

7. Pulsation: splenic arterio-venous fistula**8. Pitting sign** = Mofti's sign:

- in chronic myeloid leukemia
- Press on the surface of spleen by your thumb for a while then release the pressure, if you found an indentation, so it's positive

9. Notch in the antro-medial border

- Embryologically, the spleen is formed by fusion of splenules. The notch is formed at the site of their fusion

► Example:

Intra-abdominal swelling moves up & down with respiration, oblong in shape in left hypochondrium extending to: Felt cm (or finger breadth) below left costal margin in left mid clavicular line or Huge i.e. crossing midline, Not warm, not tender, not pulsatile, Firm in consistency, sharp anterior border with smooth surface. Hand can't reach the upper pole as you can't insinuate fingers between costal margin

► Causes of Splenomegaly :**Infectious:**

- Paratyphoid
- Typhus
- TB
- Anthrax
- Chronic malaria
- Psittacosis

Blood:

- Thrombocytopenia
- Thalassemia major

Circulatory:

- Portal vein occlusion:
- Tumour
- Thrombophlebitis

Conginital:

- Cysts of spleen

Metabolic:

- Gaucher's disease
- Porphyria
- Rickets

Tumors:

- Lymphangioma
- Fibrosarcoma
- Waldenstrom
- macroglobulinemia

► **Causes of HepatoSplenomegaly :****Infectious:**

- Typhoid
- Brucellosis
- Syphilis
- abscess
- Bilharziasis
- Hydatid cyst
- Kala azar
- Infectious mononucleosis

Blood:

- Chronic myeloid leukemia
- Hemolytic anemias
- Polycythemia rubra vera (PRV)
- Myelofibrosis

Circulatory:

- Portal hypertension

Colagen disease:

- Felty's syndrome
- Still's disease
- Sarcoidosis

Metabolic:

- Amyloidosis

Tumors:

- Hemangioma
- Malignant lymphoma

► **Important notes on liver & spleen :****Causes of tender splenomegaly: TIBS**

- Typhoid, typhus, TB (military)
- Infective endocarditis, Infectious mononucleosis, Infarction
- Brucellosis
- Septicemia

Causes of huge splenomegaly: spleen which crosses the midline:

- Portal hypertension (bilharziasis)
- Thalassemia major
- Chronic myeloid leukemia
- Chronic malaria
- Myeloproliferative disorders (Polycythaemia rubra vera & Myelofibrosis)
- Kala azar
- Gaucher's disease
- Splenic sarcoma

Why does enlarged spleen descend obliquely towards the umbilicus?

Towards right iliac fossa : due to presence of phrenicocolic ligament & it's axis.

Spleen may be found enlarged in Lt. iliac fossa in the following cases:

- Absent phrenicocolic ligament (embryological)
- Resection: hemicolectomy
- Infiltration of phrenicocolic ligament by malignancy
- Weak ligament: visceroptosis
- Huge spleen

What are the causes of palpable liver without hepatomegaly ?

Emphysema, subphrenic abscess

Causes of enlarged tender liver:

- Malignancy
- Infection: amebic hepatitis & abscess, viral hepatitis, pyogenic abscess
- Congestion: heart failure, constrictive pericarditis

Causes of nodular liver:

- Bilharziasis with coarse nodularity
- Post necrotic cirrhosis (post hepatitis)
- Malignancy
- Syphilis Gumma (hepar lobatum)
- Hydatid disease

Causes of absent splenic notch:

Congenital, infarction, adhesions, tumors.

Causes of multiple splenic notch:

Congenital, multiple infarction of the spleen

Stages of Bilharziasis:

Stage 1: Hepatomegaly

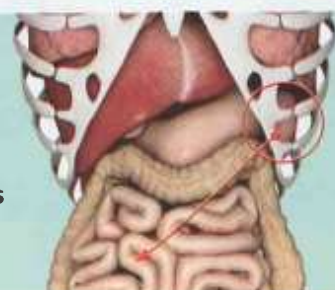
Stage 2: Hepatosplenomegaly

Stage 3: shrunken liver & splenomegaly

Stage 4: Shrunken liver, splenomegaly & ascites



My name is **Spleen**, and I enlarges that way (towards Rt. iliac) , because that guy named **Phrenicocolic ligament** prevents me from descending to the Lt. iliac fossa. You know what, If some one killed that bad boy, I would enlarges in the Lt. iliac fossa...



Palpation of Kidney

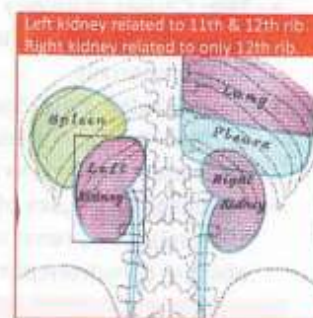
Surface anatomy of the kidney:

posteriorly the upper edge of Rt. kidney opposite the 11th dorsal spine and the lower edge of the 11th rib. Its lower edge is opposite the upper edges of L3 spine and vertebral body and about 4 cm. The left kidney is usually 1.25 cm higher, but being a little longer than the right, its lower limit may not be quite that much higher. The kidney is slightly lower in women and children than in men. The inner border reaches 10 cm. and the hilum 4 to 5 cm.

Renal Angle: Angle between last rib and lateral border of sacrospinalis muscle

Loin: lies between last rib and iliac crest and from vertebral column to flank

Flank: lateral aspect of loin



► Renal angle examination :

	Normally	Abnormally
Inspection	Skin: Normal	Skin: Ulcers, scars
	Shape :concave (Empty)	Shape: Fullness: swelling within renal capsule Bulge: swelling outside renal capsule
Palpation	No swelling	Swelling
	No tenderness	Tenderness: Thumb pressure test Costo-vertebral percussion Murphy's kidney punch
Percussion	Resonant	Dullness
Ausultation	Nothing	Bruit maybe heard in case of renal artery stenosis (Can be heard commonly anteriorly at transpyloric plane)



► Kidney Palpation :

Bimanual method:

Rt. Kidney:

The Right hand is placed anteriorly in the Right lumbar region while the Left hand is placed posteriorly in the right loin

Lt. Kidney:

The Left hand is placed anteriorly in the left lumbar region while the Right hand is placed posteriorly in the left loin.

Ballotement for examination of renal mass:

at the same previous position, push by the anterior hand and feel by the posterior hand (posterior ballotement) or push by the posterior and feel by the anterior (anterior ballotement)

Additional methods:

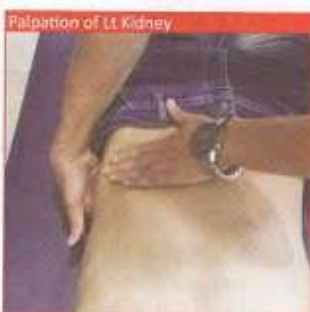
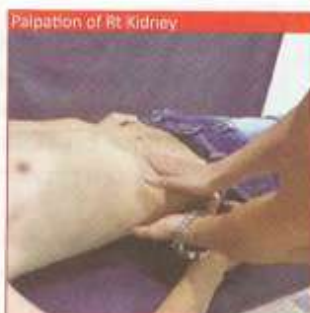
Thumb pressure test: apply pressure with the thumb to Costovertebral angle

Costovertebral percussion: using ulnar aspect of the hand strike the patient flank.

Murphy's kidney punch:

While the patient is sitting, examine for tenderness in the renal angle by using the ulnar border of the right hand (making a fist) percuss over the dorsum of the left hand placed on the renal angle; with comparing both sides.

It may help distinction between renal tenderness or tender abdominal wall or muscles of the back



Comment on following items:

- **Site:** In lumbar region
- **Size:** Causes of kidney swellings : hypernephroma, polycystic kidney, perinephric abscess & hydronephrosis
- **Border, edge:** Rounded
- **Surface:** bossy, irregular in malignancy
- **Consistency:** Cystic in polycystic kidney
- **Tenderness:** Causes of tender renal swellings: Pyelonephritis, renal infarction , perinephric abscess & hydronephrosis

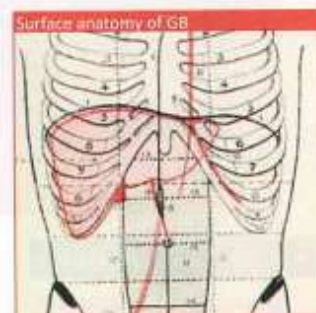
► N.B.:

Normal kidney may be palpable in: very thin persons

	Kidney	Spleen
Notch	No	Yes
Anterior & Posterior Ballotement	Ballotable (anterior & posterior)	Non ballotable Anterior only
Anterior	I can insinuate my hand between the swelling & the costal margin & I can reach the upper border	It is difficult to insinuate my hand between the swelling & the costal margin & if so I can't reach the upper border
Posterior	Can be pushed to renal angle	Can't be pushed to renal angle
Percussion	Band of resonance	Dull all through
Surface	Bossy	Smooth
Edge	Rounded	Sharp

Palpation of Gall Bladder**Anatomy**

- **Shape:** pyriform-shaped.
- **Size:** 8 - 12 cm length X 3 cm width.
- **Site:** fossa for gall bladder on the inferior surface of the right lobe of the liver.
- **Capacity:** 30 - 50 ml.
- **Power of concentration:** 10 times.
- **Parts:** funds, body and neck.



Normally GB cannot be felt, felt only if enlarged

Examination of GB is palpated for swelling& tenderness**1. Tenderness:**

Murphy's sign acute cholecystitis:

Compress your right fingertips below the right costal margin at the site of the GB & ask the patient to take deep inspiration & observe his face & breathing:

- Continue breathing: normal
- Pain & can't continue inspiration: cholecystitis

**2. Swelling:**

Site: Rt. hypochondrium

Method: Single handed method by deep palpation as liver

Comment on gall bladder swelling:

Intra-abdominal, moves up & down with respiration, in Rt. hypochondrium, pyriform or rounded, cystic or hard mass (hard in malignancy)

Painless swelling of gall bladder:

- Tumors of GB
- Cancer head of pancreas
- Courvoisier's sign: Palpable distended gall bladder in pancreatic cancer

Painfull swelling of gall bladder:

- Cystic duct stone: mucocele, pyocele
- torsion
- pericholecystic abscess

DD: Riedel's lobe (accessory hepatic lobe)

Palpation of Urinary Bladder

Normally UB is not palpable, felt only in retention.

Site: middle line, never cross umbilicus

Method:

Deep palpation from the umbilicus downwards by tips of fingers or by radial aspect of index finger of left hand

Comment distended UB "retention of urine":

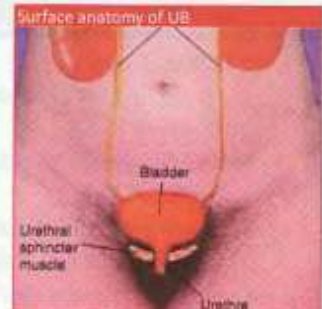
Pelvi-abdominal swelling i.e. Has no lower border

Oval in shape, smooth, firm, regular.

Direct pressure on it produces a desire to micturate, Immobile, in suprapubic region, Positive fluid thrill, Dull on percussion

Differential Diagnosis:

- Gravid uterus
- Ovarian cyst
- Any uterine mass e.g.-mesenteric cyst



Palpation of Colon

Method:

Method: Using both hands on each other (feel by the dominant hand and press by the non dominant hand) rolling method in perpendicular directions (Horizontal & Vertical)

Cecum:

Site rolling technique in right iliac fossa

Normally Cecum may be felt

Characters of Cecum:

Soft, rounded swelling with indistinct borders

Differential diagnosis: Tumors of Cecum

Sigmoid:

Site: rolling technique in left iliac fossa

Normally the pelvic colon may be palpable in thin patient or irritable bowel syndrome (spastic colon)

Characters of pelvic colon:

Oblong sausage-shaped (about 12 cm length)

Does not move with respiration, parallel to the inguinal ligament

Movable from side to side (but not vertically)

Gurgling sensation on pressure

Differential Diagnosis; mass in left iliac fossa:

Commonest in children iliac adenitis

Commonest in middle age : Bilharzial pericolic mass

Commonest in old age: cancer colon



Palpation of the Abdominal Masses

► Palpation of the Bilharzial pericolic mass :

Site: Lt. iliac fossa

Method: rolling technique

Characters:

Moves from side to side but not up down, Oval in shape, Firm & may be tender.

Pathogenesis:

heavy deposition of ova in the subserous area → Bilharzial reaction → large subserous mass (pericolic mass)

If deposition of ova submucous Bilharzial reaction intestinal polypi

Clinically the mass can't be felt by per-rectal examination

No need for excision of the mass because: It is not precancerous & does not cause obstruction

Significance: Associated with bilharzial polyposis:

DD: mass in left iliac fossa

► Palpation of Para Aortic Lymph node :

Method: Rolling technique

Site: umbilical & epigastric regions along lateral border of aorta i.e. on the left of a line extending from xiphisternum to umbilicus

Comment on: site, size, shape,

Significance: if felt in malignant cases, signifies preoperative inoperability

► Palpation of abdominal aorta :

Technique: by deep palpation using two handed method to the midline above & lateral to umbilicus.

Significance: Aortic anurysm

► Per Rectal Examination (PR) :

Patient position: left lateral position (best), Knee-elbow position

Inspection: separate buttocks carefully inspect perianal area and anus for:

- Pruritis ani (signs of External piles or fistula or inflammation)
- Anal warts, fissure
- External Haemorrhoids

Palpation:

- Put a lubricant on gloved index finger of Right hand
- Ask patient to relax & introduce your finger gently & Rotate finger 360° in anal canal searching for:
 - Thickening or irregularity in wall
 - Tone of anal musculature & sphincters
- Then pass finger into rectum with left hand placed on patient Right hip & latter in suprapubic position
- On withdrawing finger look at it for blood, mucous, pus

► N.B.: Palpation of the ascites:

Transmitted thrill (fluid thrill) in tense ascites see ascites

Value of PR:

- **Sphincter:** Loss of tone and patulous. (Cauda equina syndrome).
- **Contents:** Hard impacted stools, Foreign body.
- **Rectal wall:** Pelvic masses (Ovary, Uterus) in women.
- **Mucous membrane:** Irregular, Mass. (Cancer), piles (complicated).
- **Prostate in men:**
 - Smooth, large, firm and non-tender. (Benign enlargement)
 - Hard, irregular nodule or fixed hard mass. (Cancer, Stone, Chronic prostatitis).
 - Large, boggy and tender. (Acute Prostatitis).
- **Stools:**
 - Bloody: (Hemorrhoids, Bleeding rectal lesion).
 - Black: (Upper GI bleed, Iron, Some Antacids).



It's not funny at all. **PR** in internal medicine is **not Public Relationship**; it's **Per Rectal** examination dear...



Percussion

Simple roles:

- The percussing finger: middle finger of the right hand.
- The movement: from the wrist joint.
- The percussed finger: middle finger of the left hand

Abdominal percussion:

1. Ascites.
2. Liver borders: liver span.
3. Spleen: for splenomegaly.
4. Kidneys.
5. Bladder: for enlarged bladder or pelvic mass.
6. Masses.

► **Percussion For Ascites :**

Severe or tense	Moderate	Mild
Transmitted thrill	Shifting dullness	Knee elbow position
> 3000	1500 - 3000	50 - 1500

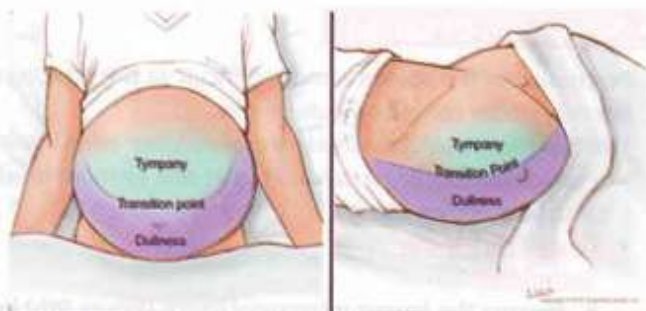
Technique:First step:

- Direct percussion is done over the abdomen, from the umbilicus to the flanks.
- The location of transition from tympany to dullness is noted.

Usually in ascites:

percussion note is tympanitic over the umbilicus and dull over the lateral abdomen and flank areas (**U shape dullness**).

The tympany over the umbilicus occurs in mild or moderate ascites, because bowel floats to the top of the abdominal fluid.

Shifting dullness:

- Percuss from the midline to the flanks on both sides:
Place the finger parallel to the expected edge. Percuss from resonance in the mid-abdomen to dullness in the flanks.
- The patient rolled to the opposite side and wait for 30 seconds.
- The previous area of dullness should now be resonant.
- It does not matter which side one chooses to start with.

If the percussion of all abdomen is dull and no tympany, use transmitted thrill to examine ascites...

Transmitted thrill (fluid thrill)

- Place one hand on the patient flanks.
- With the other hand briskly tap the other flank.
- A third hand (Patient hand) is placed in the mid-abdomen with sufficient pressure applied to dampen any wave that may pass through the anterior abdominal wall.
- **Positive test:** a shock wave be felt with palpating hand.

If the percussion of the abdomen is resonant all over the abdomen and failed to detect minimal ascites use the following technique:

Knee elbow position

Place the patient in knee elbow position and percuss the umbilicus area.



► **Percussion Of Liver :**

Aim: For detection of the liver span

Liver span:

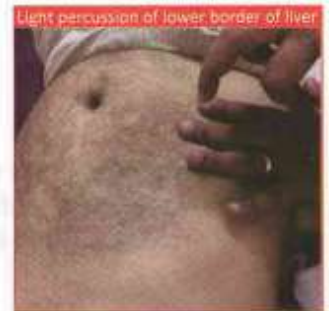
The distance between both upper and lower borders of the liver in the right MCL.
About **8 - 12 cm in a normal adult**.

The upper border (heavy percussion):

- Percuss downward from the chest in the right mid clavicular line until you detect the top edge of liver dullness. (See cardiology)
- The upper border of the liver is normally in the right 5th intercostal space.

The lower border:

- By **palpation** (see before)
- **light percussion:** Percuss upward from the abdomen in the same line until you detect the bottom edge of liver dullness.
- An alternate method of estimating liver size is the **scratch test:** Put the diaphragm of stethoscope just below Rt. Costal margin in MCL, by a sharp object scratch from the Rt. iliac fossa upwards till you hear the sound way much loader.

► **Percussion Of Spleen :**

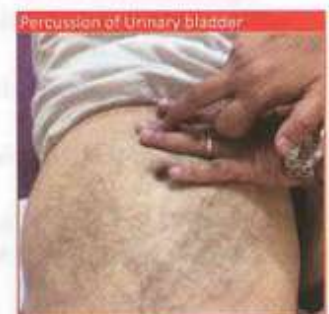
- Percussion of the spleen is more difficult as this structure is smaller and lies quite laterally under completely under ribs.
- The two most useful methods are percussion over Traub's area and Castell's sign.
- Percussion should be carried out at one or more levels of Traub's area from medial to lateral (See Chest)
- **Castell's method:**

Technique:

- Percuss the lowest intercostal space (8th or 9th) in the left anterior axillary line. With the patient in full inspiration and then full expiration.
- **Positive sign:** if the note changed from resonant on full expiration to dull on full inspiration.
- **Explanation:** (when the patient inspires, the spleen moves inferiorly along the posterolateral abdominal wall. If the spleen is enlarged enough that the inferior pole reaches the 8th or 9th intercostal space).

► **Percussion Of Urinary Bladder :**

- Percuss the upper border of the urinary bladder (Just above symphysis pubis)
- In 3 longitudinal lines (the midline and 2 para-median lines).
- The upper border of the urinary bladder (or any pelvic swelling) noticed as convex (n shape) line.



Auscultation

Auscultation of the abdomen has a relatively minor role and used for:

1. Intestinal sounds.
2. Vascular sounds.
3. Scratch test.
4. Succussion splash.
5. Buddle sign.
6. Friction rubs.



1. Intestinal sounds:

The diaphragm of the stethoscope should be placed on the abdomen, as least initially in the right lower quadrant near the ileocecal valve. Specifically listen for the pitch & frequency of the bowel sounds.

- **Normal:** gurgling, 5 - 35 per minute.
- **Rushing (Borborygmi):** loud and easily audible - normal in diarrhea.
- **Tinkling:** high pitched, (raindrops in a barrel); **intestinal obstruction.**
- **Decrease sounds:** (none for a minute) decrease gut activity. After abdominal surgery, abdominal infection (peritonitis) or injury.
- **Absent sounds:** no sounds for 3 minutes: ileus.

2. Vascular sound:

	Arterial bruits	Venous hum
Timing	Only during systole	Continuous
By	Diaphragm	Bell
Characterized by	High - pitched	Low - pitched
Increased by	---	Inspiration

• Venous hum:

Best heard overlying the portal vein in an area approximated by an elliptical shape between the umbilicus and the mid-clavicular line where it crosses the right subcostal margin.

• Arterial bruits:

Aorta	Begins at or just to the left of the xiphoid process and bifurcates at the level of the umbilicus.
Renal a.	Midway between the xiphoid process and the umbilicus and within 2 cm of the midline.
Bifurcation of CIA	Mid-way between the umbilicus and the mid inguinal point (MIP).
Femoral	Just above the inguinal ligament (MIP)

3. Scratch test:

Technique:

- Place the diaphragm over the area of liver/spleen
- Then scratch parallel to the costal margin until the sound intensity drops of marking the edge of the liver.
- Different patterns of scratch may be used (e.g. spokes of a wheel). see picture before

► N.B.:

Blood flow through the aorta itself usually does not generate any appreciable sound.

4. Succussion splash:

- Called auscultation of the stomach
- Detected in gastric outlet obstruction e.g. pyloric ulcer or neoplasm.
- **Technique:**
 - Place the stethoscope on the epigastrium.
 - Then shake both iliac crests.
 - While shaking, listen to splash from retained fluid.

5. Buddle sign:

- In knee elbow position.
- It is useful for detecting small amounts of ascites as small as 120 ml (shifting dullness and bulging flanks typically requires 500 ml).

6. Friction rubs (rare)

- Right and left upper quadrant.
- Grating sound with respiratory movement.
- Indicates inflammation of peritoneal surface of an organ.

► **N.B.:****Auscultation over the liver:**

- Friction rub (grating during breathing) peritonitis.
- Bruit (cancer).
- Scratch.

Auscultation is sometimes done before percussion and palpation, unlike in other examination.

It may be performed first because vigorously touching the abdomen may disturb the intestines, perhaps artificially altering their activity and thus the bowel sounds.

Additionally, it is least likely to be painful - invasive; if the person has peritonitis and you check for rebound tenderness and then want to auscultate you may no longer have a cooperative patient.

LONG ABDOMINAL CASE

► **History :**

Personal history: Female patient, xxx, 47 years old, from Al-Fayoum, house wife, married 29 years ago and has 3 sons, the youngest is 19 years old, she has no special habits of medical importance.

Complaint: She is complaining of abdominal swelling of 2 weeks duration.

History of present illness:

The condition started 15 years ago by an active attack of haematemesis of sudden onset with 2 cups of dark brown blood preceded by nausea and followed by melena for 3 days, she admitted in Aldemerdash hospital, investigated by upper endoscopy & treated by endoscopic sclerotherapy without blood transfusion and this attack is not recurrent until now.

- No appetite changes, dysphagia, heart burn or halitosis.
- No changes in bowel habits or bleeding per rectum.
- 10 years ago, patient suffered from lower limbs oedema with gradual onset and progressive course, responds to diuretics.
- Followed after 2 years by ascites with gradual onset and progressive course with several paracentesis process of about 3 liters of clear fluid and not followed by any complications.
- Lower limb oedema & ascites are associated with jaundice and pruritis with dark urine and low grade fever & progressive mild loss of weight.
- No abdominal pain or urinary symptoms.
- No gynaecological symptoms.
- No symptoms of other system affection.
- Patient is investigated by CBC, liver function tests and abdominal ultrasonography then treated by vitamins & diuretics.

Past history:

History of bilharziasis at age of 13, manifested by dysentery, investigated by urine & stool analysis and treated by praziquantil.

No history of other chronic diseases, surgical operations or drug intake.

Family history: Irrelevant.

► **General Examination :**

- Patient with an average general condition, average built.
- Patient is fully conscious, oriented for time, place and persons
- Patient lies in semi sitting position.

Head and face examination:

- Orange yellow jaundice, wasted temporalis, with endemic parotitis.
- No pallor or cyanosis.
- No puffy eyelids with normal eye brows.
- Bad oral hygiene with normal tongue size.

Neck examination:

- Congested pulsating neck veins, JVP = 4 cm above sternal angle with normal waves.
- Palpable carotid pulsation without thrill.
- Central trachea - no lymph node enlargement or thyroid swelling.
- Spider nevus in root of the neck.

Upper limb examination:

- Pulse: 75 beats / minute, regular, big volume,
- Blood pressure: 130 / 50 mmHg.
- Palmar erythema: are present in both hands.

Lower limbs examination:

- Shows bilateral, painless, pitting oedema.
- Echymotic patches: are present in lower limbs, upper limbs and trunk.
- Systems examination: no abnormalities were detected are regard C.V.S., chest & C.N.S. examination.

► **Local Examination :****Inspection:**

- Symmetrical enlargement of the abdomen with full flanks, wide subcostal angle, and abdomen moves freely with respiration.
- No gynecomastia, epigastric pulsation.
- There are divercation of recti, umbilicus is shifted downward, everted with umbilical hernia, no pigmentation, discharge or nodules.
- No striae, scar or hernia neither dilated veins.
- No swelling in renal angle or back deformity.

Palpation:**A. Superficial**

No superficial swelling or tenderness.

B. Deep palpation

- Right & lower lobes of liver are not palpable.
- Spleen: swelling in the left hypochondrium intra-abdominal, 5 fingers below the left costal margin with notch, sharp edge firm in consistency, smooth surface, not tender & not pulsating.
- Kidney are not palpable.
- No any palpable mass.

Percussion:

- Moderate ascites is detected by shifting dullness.
- Dullness on Traub's area.

Auscultation:

- Normal intestinal sounds.
- No vascular sounds, rubs or splachs.

► **Proper Diagnosis :**

A case of shrunked liver with splenomegaly for D.D. most probably due to mixed bilharzial & hepatitis cirrhosis, with vascular and cellular decompensation.